

Chapter 2 - BASIC Language Vocabulary

This chapter is a reference to every word in BASIC. The words are listed in alphabetical order. Each entry shows the syntax form, a short description, and an example. Hardware-related keywords give the chapter where the screen, sound, or input system they belong to is described in full.

Do not read this chapter as a lesson. On a first pass, skim the headings so you know what BASIC can say, then continue to Chapter 3 and make the machine draw. Return here when you need the exact spelling of a keyword or the argument order for a command. BASIC is the common door into the same bus machine used by the later graphics, sound, file, and processor chapters.

2.1 Notation

The following conventions appear throughout this chapter and the rest of the book:

Notation	Meaning
UPPERCASE	Type this word exactly.
<i>italics</i>	A piece of variable information you supply.
[...]	Optional. May be omitted.
[...]*	Optional and may be repeated.
a \ b	One of a or b.
<i>expr</i>	Any numeric expression.
<i>str-expr</i>	Any string expression.
<i>var</i>	A numeric variable name.
<i>str-var</i>	A string variable name (ends with \$).
<i>line</i>	A line number (0-4294967295).
<i>addr</i>	An integer expression interpreted as an address.

2.2 Direct-mode commands

These commands work only in direct mode. They are not part of the stored program language and may not appear inside a stored program line.

- **DIR** - list the names of files that BASIC can LOAD. See Chapter 35.
- **RUN AOT** - compile the stored program to native IE64 code and run it immediately.
- **COMPILE** - write the stored program as a standalone .ie64 image. See Chapter 35.
- **TRANSPILE** - write the stored program as IE64 assembly text. See Chapter 35.
- **ASSEMBLE** - assemble IE64 source text into a standalone .ie64 image. See Chapter 35.

2.3 Alphabetical reference

ABS

ABS (expr)

Returns the absolute value of *expr*.

```
PRINT ABS ( -5.5 )  
5.5
```

AHX

AHXargs

Play an AHX music file. See Chapter 18.

AND

exprANDexpr

Bitwise AND of two integers. Each operand is truncated to a signed 32-bit integer before the operation; the result is converted back to a number.

```
PRINT 12 AND 10  
8
```

ANTIC

ANTICargs

Program the ANTIC display list. See Chapter 7.

ASC

ASC (str-expr)

Returns the byte value of the first character of *str-expr*. If the string is empty, returns 0. See Appendix C for the character code table.

```
PRINT ASC ( "A" )  
65
```

ASSEMBLE

ASSEMBLE "filename"

Direct-mode command. Read the assembly source matching *filename* from the disk volume, assemble it inside the machine as IE64 source at `PROG_START`, and write *filename.ie64*. The stored BASIC program is not changed.

ASSEMBLE accepts the IE64 instruction set, labels, branch and call targets, `dc.b`, `dc.w`, `dc.l`, `dc.q`, `align`, and the symbolic constants known to the IE64 assembler. See Chapter 35 for the file behaviour and Chapter 25 for the IE64 image form.

ATN

`ATN(expr)`

Returns the arctangent of *expr* in radians.

BIN\$

`BIN$(expr)`

Returns the binary representation of the integer part of *expr* as a string of 0 and 1 digits.

```
PRINT BIN$(10)
1010
```

BITCLR

`BITCLRaddr,bit`

Clears bit *bit* (0-7) of the byte at *addr*. *bit* outside that range gives an unspecified result.

BITSET

`BITSETaddr,bit`

Sets bit *bit* (0-7) of the byte at *addr*.

BITTST

`BITTST(addr,bit)`

Returns -1 if bit *bit* (0-7) of the byte at *addr* is set, 0 if it is clear.

BLIT

`BLIT COPYsrc,dst,w,h [, src-stride,dst-stride]`

`BLIT FILLdst,w,h,colour [, dst-stride]`

`BLIT LINEx0,y0,x1,y1,colour [, dst-stride]`

`BLIT MEMCOPYsrc,dst,len`

`BLIT Msrc,dst,len`

`BLIT MODE7src,dst,w,h,u0,v0,du-col,dv-col,du-row,dv-row,u-mask,v-mask [, src-stride,dst-stride]`

`BLIT WAIT`

Use the VideoChip blitter. COPY copies pixels, FILL fills a rectangle, LINE draws one line, MEMCOPY copies a byte span, M is the short form of MEMCOPY, MODE7 performs affine texture mapping, and WAIT waits until the blitter is idle. See Chapter 4.

BLOAD

`BLOADfilename,addr`

Load a binary file into memory beginning at *addr*. Both arguments are required; omitting the comma and address loads nothing. The destination is carried through the File I/O block's 32-bit data pointer; an address of 2³² or greater reports ? FC ERROR. See Chapter 35.

BOX

`BOXx0,y0,x1,y1[,colour]`

Draw a rectangle. See Chapter 5.

CALL

`CALLaddr`

Call IE64 machine code at *addr*. Returns to BASIC when the called code executes RTS. See Chapter 25 for the calling convention.

CHR\$

`CHR$(expr)`

Returns a one-character string whose single byte equals the low eight bits of `INT(expr)`. `CHR$(n)` and `CHR$(n+256)` produce the same string.

```
PRINT CHR$(65); CHR$(66); CHR$(67)
ABC
```

CIRCLE

`CIRCLEx,y,r[,colour]`

Draw a circle. See Chapter 5.

CLEAR

`CLEAR`

Reset all variables and arrays. Reset the DATA read pointer. Reset the GOSUB and structured-control stack. The program text is preserved.

CLS

`CLS`

Clear the screen. See Chapter 5.

COMPILE

`COMPILE "filename"`

Direct-mode command. Compile the stored BASIC program into a standalone IE64 image. If *filename* has no `.ie64` suffix, BASIC adds it. If the current program was loaded from a subdirectory, the compiled image is written beside that loaded program; otherwise it is written at the root of the disk volume.

COMPILE is a prompt command, not a stored-program statement. It rejects direct-mode-only commands and other raw prompt forms. When a statement cannot be compiled, BASIC prints `?COMPILE ERROR IN` followed by the line number and the reason. See Chapter 35 for the File I/O side and Chapter 25 for flat IE64 images.

COLOR

COLOR fg [, bg]

Set the VGA text-mode foreground and optional background colour used by PRINT. See Chapter 5.

CONT

CONT

Resume execution after a STOP. Has no effect if there is no saved state. After RUN AOT, a top-level STOP saves a native IE64 continuation, and CONT re-enters the compiled code. Editing the program, NEW, LOAD, or a fresh RUN or RUN AOT discards that saved continuation.

COPPER

COPPER $args$

Program the copper list. See Chapter 4.

COS

COS($expr$)

Returns the cosine of $expr$, where $expr$ is in radians.

COSTART

COSTART $args$

Start a coprocessor program. See Chapter 32.

COSTOP

COSTOP $args$

Stop a coprocessor program. See Chapter 32.

COWAIT

COWAIT $args$

Wait for a coprocessor to reach a synchronisation point. See Chapter 32.

DATA

DATA $item$ [, $item$]*

Declare numeric or string constants for later use with READ. The items are stored as raw text from DATA to the end of the line or to the next colon, and parsed when READ consumes them.

```
10 DATA 1, 2, "HELLO", 3.5
20 READ A, B, C$, D
```

DEC

DEC var

Subtract 1 from *var*. Equivalent to `LET var=var- 1`, but shorter to type.

DEEK

`DEEK(addr)`

Read 16 bits at *addr* (unsigned). The high 16 bits of the result are zero.

DEF

`DEF FNname(var) =expr`

Define a user function with one parameter. The function is called later with `FNname(expr)`. Use the exact `DEF FN` form for user functions.

NOTE: `DEF` shares stored token byte \$97 with `TROFF`. Type `DEF FN` exactly. If you `LIST` a stored function definition, the listing may show `TROFF` in place of `DEF`; the line still behaves as `DEF FN`. Appendix A records the token map.

DIM

`DIMname(size) [, name(size[, size])]`

Declare a numeric array. *size* is the highest legal subscript; the array has *size* + 1 elements per dimension, indexed from 0. Only numeric arrays are supported. There is no string-array form.

```
DIM A(10), GRID(7,7)
```

DO

`DO`

Begin a structured loop. End the loop with `LOOP UNTILcondition` or `LOOP WHILEcondition`. See **LOOP**.

DOKE

`DOKEaddr, expr`

Write 16 bits at *addr*. *addr* may be any byte address; alignment is not required.

ELSE

`IFcondition THEN then-part ELSE else-part`

Mark the alternative branch of an `IF` statement. See **IF**.

END

`END`

Stop the running program and return to direct mode. There is no saved state for `CONT`.

ENVELOPE

`ENVELOPEargs`

Set a volume envelope. See Chapter 11.

EOR

*expr*EOR*expr*

Bitwise exclusive-OR of two integers.

```
PRINT 12 EOR 10
6
```

EXP

EXP(*expr*)

Returns *e* raised to the power *expr*.

FN

FN*name*(*expr*)

Call a user function defined with DEF FN.

FOR

FOR*var*=*start*TO*end*[STEP*step*]

Begin a numeric loop. *var* takes the value *start* on first entry. After each pass through the loop body, the matching NEXT adds *step* (default 1) to *var* and compares it with *end*. The loop ends when *var* passes *end*.

```
10 FOR I = 1 TO 10
20   PRINT I,
30 NEXT
40 PRINT
```

FRE

FRE(*expr*)

Returns the approximate number of bytes of free string heap. The argument is ignored; use FRE(0).

GATE

GATE*args*

Open or close a sound-channel gate. See Chapter 11.

GET

GET*var* | GET*str-var*

Read one character from the keyboard without waiting for RETURN. *var* receives the byte value; *str-var* receives the character as a one-byte string. If no key is available, *var* is 0 and *str-var* is the empty string.

GOSUB

GOSUB*line*

Call the subroutine starting at *line*. Execution resumes at the statement after the matching RETURN. Subroutines may call other subroutines.

GOTO

GOTO*line*

Branch to *line*. Execution continues at the first statement of that line.

GTIA

GTIA*args*

Program the GTIA. See Chapter 7.

HEX\$

HEX\$(*expr*)

Returns the hexadecimal representation of INT(*expr*) as a string of uppercase hex digits.

```
PRINT HEX$(255)
FF
```

HOST

HOST*subverb*

Issue a HOST command. *subverb* is one of NET, UPDATE, REBOOT, POWEROFF, HELP. See Chapter 36.

IF

IF*condition*THEN*then-part*[ELSE*else-part*]

If *condition* is non-zero, execute *then-part*; otherwise execute *else-part* if present, or fall through. *then-part* may be a line number (equivalent to GOTO*line*) or a statement list.

```
10 INPUT N
20 IF N < 0 THEN PRINT "MINUS" ELSE PRINT "PLUS OR ZERO"
```

INC

INC*var*

Add 1 to *var*.

INPUT

INPUT [*prompt*;]*var* [, *var*]*

Print *prompt* (a quoted string) if present, then ?, then read values typed on the terminal. Values are separated by commas.

```
10 INPUT "WHAT IS YOUR NAME"; N$
20 PRINT "HELLO, "; N$
```

INT

INT(*expr*)

Returns the integer part of *expr*, truncated toward 0.

LCASE\$

LCASE\$(*str-expr*)

Returns *str-expr* with every uppercase letter replaced by its lowercase equivalent. Non-letter bytes are unchanged.

LEEK

LEEK(*addr*)

Read 32 bits at *addr*. Equivalent to PEEK(*addr*).

LEFT\$

LEFT\$(*str-expr*, *n*)

Returns the first *n* bytes of *str-expr*. If *n* is negative, raises ILLEGAL QUANTITY.

LEN

LEN(*str-expr*)

Returns the number of bytes in *str-expr*.

LET

[LET] *var*=*expr*

Assign *expr* to *var*. The LET keyword is optional.

LINE

LINE *x0*, *y0*, *x1*, *y1* [, *colour*]

Draw a line. See Chapter 5.

LIST

LIST

Print the whole stored program. Any arguments after LIST are ignored; BASIC always lists every line.

LOAD

LOAD *filename*

Load a program file. See Chapter 35.

LOCATE

LOCATE *row*, *col*

Move the text cursor. See Chapter 5.

LOG

LOG(*expr*)

Returns the natural logarithm of *expr*. *expr* must be positive.

LOKE

LOKE*addr, expr*

Write 32 bits at *addr*. *addr* must be a multiple of 4. Equivalent to POKE*addr, expr*.

LOOP

LOOP UNTIL*condition* | LOOP WHILE*condition*

End a DO block. LOOP UNTIL repeats until *condition* is non-zero; LOOP WHILE repeats while *condition* is non-zero.

```
10 N = 0
20 DO
30   N = N + 1
40   PRINT N
50 LOOP UNTIL N >= 5
```

MAX

MAX(*expr, expr*)

Returns the larger of the two arguments.

MID\$

MID\$(*str-expr, start* [, *len*])

Returns *len* bytes of *str-expr* starting at position *start* (1-based). If *len* is omitted, returns the substring from *start* to the end.

MIN

MIN(*expr, expr*)

Returns the smaller of the two arguments.

MOD

MOD STATUS

Return the MOD player's status byte. Use SOUND MOD PLAY, SOUND MOD STOP, and SOUND MOD FILTER to control playback. See Chapter 19.

NEW

NEW

Erase the program, all variables, and all arrays. After NEW, the program is empty.

NEXT

NEXT [*var*]

Mark the end of a FOR loop. *var* identifies which loop to close; if omitted, the innermost loop is used.

NOT

NOT*expr*

Bitwise complement of INT(*expr*) as a signed 32-bit integer.

ON

ON*expr*GOTO*line1*[, *line2*]*

Branch to *line-N*, where *N* is INT(*expr*). *N* must be between 1 and the number of line numbers listed. If *N* is out of range, execution falls through.

ON*expr*GOSUB*line1*[, *line2*]*

Like the GOTO form but each branch is a subroutine call.

OR

*expr*OR*expr*

Bitwise OR of two integers.

PALETTE

PALETTE*args*

Set entries in the colour palette. See Chapter 3.

PEEK

PEEK(*addr*)

Returns the 32-bit value at *addr*. *addr* must be a multiple of 4.

PEEK8(*addr*)

Returns the byte value at *addr*. *addr* may be any byte address.

PI

The constant 3.141593. Use 2 * PI for a full turn, or use TWOPI.

PLOT

PLOT*x*,*y*[, *colour*]

Set a single pixel. See Chapter 5.

POKE

POKE*addr*,*expr*

Write 32 bits at *addr*. *addr* must be a multiple of 4. If the address is not aligned, BASIC reports ILLEGAL FUNCTION CALL.

`POKE8addr, expr`

Write the low byte of *expr* at *addr*. *addr* may be any byte address.

POKEY

`POKEYargs`

Program the POKEY sound chip. See Chapter 17.

POS

`POS(expr)`

Returns the current cursor column (0-based). The argument is ignored. Use `POS(0)`.

PRINT

`PRINT [item[,|; item]*[;]]``

Print zero or more items to the terminal. The separators have these effects:

Separator	Effect
<code>;</code>	No padding between items.
<code>,</code>	Insert one TAB byte (<code>CHR\$(9)</code>) between items.
trailing <code>;</code>	Suppress the closing newline.
trailing <code>,</code>	Insert one TAB byte; do not suppress the newline.

The single character `?` is an alternate spelling of `PRINT`. `LIST` prints it as `PRINT`.

```
PRINT "A"; "B"; "C"
ABC
```

PSG

`PSGargs`

Program the PSG sound chip. See Chapter 13.

READ

`READvar[, var]*`

Consume successive items from the program's `DATA` statements and assign them to *var*. Numeric variables get numeric items; string variables get string items. `RESTORE` resets the read pointer.

REM

`REMcomment`

Everything from `REM` to the end of the line is a comment.

RESTORE

RESTORE

Reset the DATA read pointer to the first DATA item in the program.

RETURN

RETURN

End a GOSUB. Execution resumes at the statement after the GOSUB that called this subroutine.

RIGHT\$

RIGHT\$(*str-expr*, *n*)

Returns the last *n* bytes of *str-expr*.

RND

RND(*expr*)

Returns a pseudo-random number in the range [0, 1). A negative argument seeds the generator; otherwise the argument is ignored.

RUN

RUN

Run the stored program from the lowest-numbered line. Numeric line arguments are not parsed, so RUN 100 behaves like RUN and always restarts from the first stored line. RUN preserves variables and arrays; only the DATA pointer and the control stack are reset.

RUN AOT

Direct-mode form. Compile the stored program to native IE64 code inside the machine and run the compiled code immediately. The visible program result should match RUN, but BASIC first prints:

```
Compiling to native code...
```

RUN AOT restarts from the first stored line, resets the same program state as RUN, and discards any older compiled STOP continuation.

RUN "*filename*" is a direct-mode form for running a saved program image. See Chapter 35.

SADD

SADD(*str-var*)

Returns the address of the byte buffer that holds the current value of *str-var*. The address is invalidated by the next string allocation, so save the bytes you need before BASIC creates a new string.

SAP

SAPargs

Play a SAP music file through the POKEY. See Chapter 17.

SAVE

SAVEfilename

Save the program to a file. See Chapter 35.

SCREEN

SCREENargs

Select a screen mode. See Chapter 5.

SCROLL

SCROLLargs

Scroll the VGA screen. See Chapter 5.

SGN

SGN(expr)

Returns -1, 0, or +1 depending on the sign of *expr*.

SID

SIDargs

Program the SID sound chip. See Chapter 15.

SIN

SIN(expr)

Returns the sine of *expr*, in radians.

SOUND

SOUNDargs

Play a tone on a sound channel. See Chapter 11.

SQR

SQR(expr)

Returns the square root of *expr*. *expr* must not be negative.

STEP

See **FOR**.

STOP

STOP

Stop the program and save its position so that CONT can resume. For RUN AOT, this is a native continuation inside the compiled IE64 code. A STOP reached inside active compiled GOSUB nesting is not a resumable subroutine state; CONT

resumes after the STOP, but a following RETURN reports ?RETURN WITHOUT GOSUB.

STR\$

STR\$(*expr*)

Returns the decimal string representation of *expr*. A leading space is reserved for the sign: a positive number has a leading space; a negative number has a leading -.

SWAP

SWAP *var*, *var*

Exchange the values of two variables. Both variables must be of the same type (both numeric or both string).

TAB

TAB(*expr*)

Used only inside a PRINT list. Prints spaces so that the cursor sits at column *expr* (counting from 0). Has no effect if the cursor has already passed *expr*.

TAN

TAN(*expr*)

Returns the tangent of *expr*, in radians.

TED

TED *args*

Program the TED chip (video and audio). See Chapter 6 and Chapter 16.

TEXTURE

TEXTURE *args*

Bind a texture for Voodoo rasterisation. See Chapter 9.

THEN

See **IF**.

TO

See **FOR**.

TRIANGLE

TRIANGLE *args*

Rasterise a Voodoo triangle. Use TRICOLOR in the argument list to give one colour for each vertex. See Chapter 9.

TROFF

TROFF

Turn off line-by-line tracing.

NOTE: TR0FF shares stored token byte \$97 with DEF. When the token appears before FN, BASIC treats it as a function definition. Appendix A records the token map.

TRON

TRON

Turn on line-by-line tracing. While tracing is on, each line number is printed before its statements run.

TRANSPILE

TRANSPILE *"filename"*

Direct-mode command. Convert the stored BASIC program to IE64 assembly text and write the matching assembly source file. It does not assemble the text and it does not write a .ie64 image.

The assembly text is the same self-contained source that COMPILE writes beside its image. ASSEMBLE *"filename"* can assemble it later into the same kind of flat IE64 image. See Chapter 35.

TWOPI

The constant 6.283185 (two times pi).

UCASE\$

UCASE\$(*str-expr*)

Returns *str-expr* with every lowercase letter replaced by its uppercase equivalent. Non-letter bytes are unchanged.

ULA

ULAargs

Program the ULA display. See Chapter 8.

UNTIL

See **LOOP**.

USR

USR(*addr*)

Call IE64 machine code at *addr*. The routine returns an integer value through the result register; USR returns that value as a number. See Chapter 25 for the calling convention.

VAL

VAL(*str-expr*)

Parse *str-expr* as a number. Stops at the first character that is not part of a numeric literal.

```
PRINT VAL("3.14XYZ")
3.140000
```

VARPTR

VARPTR(*var*)

Returns the address of the storage slot that holds the value of *var*. The slot is four bytes wide.

VERTEX

VERTEX*args*

Submit a Voodoo vertex. See Chapter 9.

VOODOO

VOODOO*args*

Program the Voodoo rasteriser. See Chapter 9.

VSYNC

VSYNC

Wait for vertical retrace. See Chapter 3.

WAIT

WAIT*addr*,*mask*[,*xor*]

Read the 32-bit value at *addr* repeatedly until ((*value* EOR*xor*) AND*mask*) is non-zero. *xor* defaults to 0. WAIT returns after approximately one million polls if the condition is never met.

```
REM WAIT UNTIL A COOKED KEY BYTE IS QUEUED
WAIT &H000F072C,1
PRINT PEEK(&H000F0728)
0
```

The polled byte is 0 until a key has been typed; press a key before the example runs to see the cooked code instead.

WEND

WEND

Alternate spelling of UNTIL when terminating a WHILE loop. LIST prints this form as UNTIL.

WHILE

WHILE*condition*

Begin a loop that runs while *condition* is non-zero. The loop body runs until a matching LOOP UNTIL or LOOP WHILE. See LOOP.

XOR

There is no XOR keyword in this BASIC. The exclusive-OR operator is EOR.

ZBUFFER

ZBUFFERargs

Enable or configure the Voodoo Z-buffer. See Chapter 9.

2.4 Operator symbols

Every operator is described at its alphabetical position above. The operator symbols are summarised here for quick reference. See Chapter 1 §1.8 for the precedence table.

Symbol	Keyword name
+	(add)
-	(subtract or unary minus)
*	(multiply)
/	(divide)
^	(power)
<<	LSHIFT
>>	RSHIFT
<	LT
<=	(LT followed by =)
=	EQUAL
<>	(LT followed by >)
>	GT
>=	(GT followed by =)
?	PRINT (alternate spelling)

2.5 Cross-reference index

The hardware-related keywords above point at the chapter that documents the underlying device in full. The list below collects those references:

Keyword	Chapter
SCREEN, CLS, PLOT, LINE, CIRCLE, BOX, LOCATE	5
PALETTE, VSYNC	3
COLOR	5
COPPER, BLIT	4
SCROLL	5
TED	6 (video), 16 (audio)
ANTIC, GTIA	7
ULA	8

Keyword	Chapter
VOODOO, ZBUFFER, VERTEX, TRIANGLE, TEXTURE	9
SOUND, ENVELOPE, GATE	11
SOUND PLAY, SOUND STOP	23
PSG	13
SID	15
POKEY, SAP	17
AHX	18
SOUND MOD, MOD STATUS	19
HOST	36
COSTART, COSTOP, COWAIT	32
CALL, USR	25
LOAD, SAVE, BLOAD, RUN ".ie", RUN AOT, COMPILE, TRANSPILE, ASSEMBLE	35